



Fig. 2: Leakage power in CMOS

Ways to reduce power consumption

Depending upon the architecture of the design, the designers can choose from a wide range of options for reduction in power consumption of VLSI circuits. Some of the techniques to reduce the power consumption are briefed below.

Logic restructuring. The CMOS design uses both NMOS and PMOS transistors for logic design. Due to mobility variation in NMOS and PMOS transistors, the rise and fall time differ. By moving high switching operations up in the logic cone and low switching operations back in the logic cone, a considerable amount of power can be saved.

Clock tree optimisation and clock gating. Most of the power is consumed due to the high clock frequency used for operating the device. Portions of the clock tree that are not being used at any particular time can be disabled to save the power.

Logic resizing (transistor resizing). The length-to-width ratio of transistors determines the driving strength and speed. Upsizing improves slow times, reducing dynamic current. Hence the dynamic power consumption is reduced. Downsizing reduces leakage current, thereby reducing the static power consumption. To be effective, sizing operations must include accurate switching information.

Transition rate buffering. Buffer manipulation reduces dynamic power

by minimising switching time, hence dynamic power consumption can be reduced.

Pin swapping. By swapping gate pins, switching occurs at gates/pins with lower capacitive loads.

Using multi-threshold voltage (V_{th}). The device can be scaled in terms of voltage; since the supply voltage has quadrature effect in dynamic power consumption, reducing the voltage may lead to slower operation. Hence multi-threshold voltage device concept can be used for optimising between speed and power. With the use of multi-threshold libraries, individual logic gates use transistors with low switching thresholds (faster with higher leakage) or high switching thresholds (slower with lower leakage).

Multi-supply voltage (voltage islands). The concept of multi-supply voltage can be used in different functional blocks of the core. Select functional blocks are run at different supply voltages.

Dynamic voltage scaling. In this subset of DVFS, select portions of the device are dynamically set to run at different voltages on the fly while the chip is running.

Dynamic voltage and frequency scaling (DVFS). Select portions of the device are dynamically set to run at different voltages and frequencies on the fly while the chip is running. This technique is used for dynamic power reduction.

Adaptive voltage and frequency

scaling. In this variation of DVFS, a wider variety of voltages are set dynamically based on the adaptive feedback from a control loop; it involves analogue circuitry.

Power shutoff or power gating. Most of the time many functional blocks and even the system may be in inactive state. When not in use, select functional blocks are individually powered down. The device which needs attention in the inactive mode can be put to idle or sleep mode and all the necessary power modules and clock can be gated so that power consumption is reduced.

Memory splitting. If the software and/or data are persistent in one portion of a memory but not in another, it may be appropriate to split that block of the memory into two or more portions. One can then selectively power down those portions which aren't in use.

Substrate biasing (body-biasing or back-biasing). Substrate biasing in PMOS biases the body of the transistor to a voltage higher than V_{dd} ; in NMOS, to a voltage lower than V_{ss} . This reduces the leakage current.

Operand isolation. Power dissipation can be reduced in data path blocks controlled by an enable signal; when the data path element is not active, it is prevented from switching. ●

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